A Novel Deep Learning System for Automated Diagnosis and Grading of Lumbar Spinal Stenosis Based on Spine MRI: Model Development and Validation

Authors:	Tianyi Wang, Aobo Wang, Yiling Zhang, Xingyu Liu, Ning Fan, Shuo Yuan, Peng Du, Qichao Wu, Ruiyuan Chen, Yu Xi, Zhao Gu, Qi Fei, Lei Zang
Publication / Venue:	Neurosurgical Focus
Year:	2025
DOI:	10.3171/2025.4.focus24670
Primary Source URL:	https://pubmed.ncbi.nlm.nih.gov/40591977/
Inventory Category:	DL stenosis / multi-pathology
Comparability / Priority:	Very High | Priority: Critical

Study Details & Methodology from Literature Inventory

Study Type / MRI View:	Development + validation | Lumbar MRI
Dataset & Cohort:	None (Sample: None)
Disease / Target:	Central, lateral recess and foraminal stenosis
Model / AI Method:	Deep-learning screening/grading system
Evaluation Metrics:	None
Key Finding / Relevance:	None
Thesis Context (Ch 2):	Recent direct multi-type stenosis-grading study.

Abstract / Summary

OBJECTIVE The study aimed to develop a single-stage deep learning (DL) screening system for automated binary and multiclass grading of lumbar central stenosis (LCS), lateral recess stenosis (LRS), and lumbar foraminal stenosis (LFS). METHODS Consecutive inpatients who underwent lumbar MRI at our center were retrospectively reviewed for the internal dataset. Axial and sagittal lumbar MRI scans were collected. Based on a new MRI diagnostic criterion, all MRI studies were labeled by two spine specialists and calibrated by a third spine specialist to serve as reference standard. Furthermore, two spine clinicians labeled all MRI studies independently to compare interobserver reliability with the DL model. Samples were assigned into training, validation, and test sets at a proportion of 8:1:1. Additional patients from another center were enrolled as the external test dataset. A modified single-stage YOLOv5 network was designed for simultaneous detection of regions of interest (ROIs) and grading of LCS, LRS, and LFS. Quantitative evaluation metrics of exactitude and reliability for the model were computed. RESULTS In total, 420 and 50 patients were enrolled in the internal and external datasets. High recalls of 97.4%–99.8% were achieved for ROI detection of lumbar spinal stenosis (LSS). The system revealed multigrade area under curve (AUC) values of 0.93–0.97 in the internal test set and 0.85–0.94 in the external test set for LCS, LRS, and LFS. In binary grading, the DL model achieved high sensitivities of 0.97 for LCS, 0.98 for LRS, and 0.96 for LFS, slightly better than those achieved by spine clinicians in the internal test set. In the external test set, the binary sensitivities were 0.98 for LCS, 0.96 for LRS, and 0.95 for LFS. For reliability assessment, the kappa coefficients between the DL model and reference standard were 0.92, 0.88, and 0.91 for LCS, LRS, and LFS, respectively, slightly higher than those evaluated by nonexpert spine clinicians. CONCLUSIONS The authors designed a novel DL system that demonstrated promising performance, especially in sensitivity, for automated diagnosis and grading of different types of lumbar spinal stenosis using spine MRI. The reliability of the system was better than that of spine surgeons. The authors' system may serve as a triage tool for LSS to reduce misdiagnosis and optimize routine processes in clinical work.